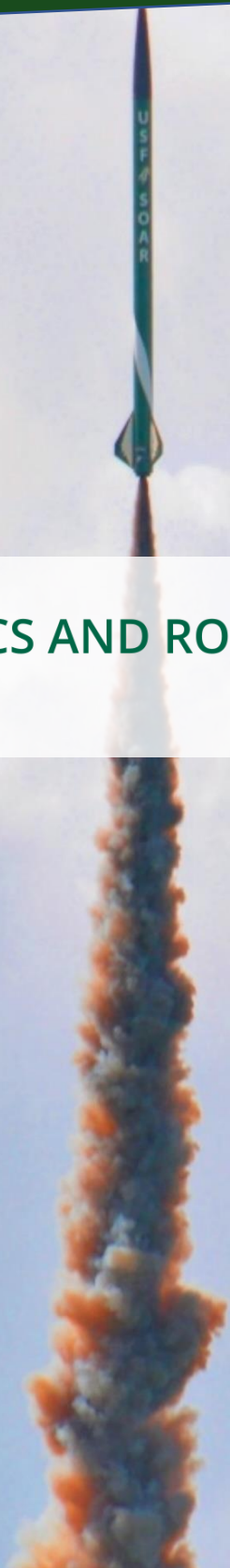


2024-2025



USF SOCIETY OF AERONAUTICS AND ROCKETRY
THE SKY IS NOT THE LIMIT.
Sponsorship Packet



Who are we?

The Society of Aeronautics and Rocketry (SOAR) at the University of South Florida is a student-led, non-profit organization established in 2013 by a group of passionate and driven undergraduate engineering students. Our organization's main operations include the design and manufacturing of high-powered rockets annually to carry out missions and/or experiments to compete in the NASA University Student Launch Initiative (USLI, also known as NSL). Our main operations also include scientific research efforts over the past years to develop a liquid rocket engine called Torito. SOAR also takes on projects that advance students' technical skills, organizational knowledge, and professional development. All of these competitions, projects, and skillsets are detailed on the following pages.

We provide opportunities for students across all majors, backgrounds, and campuses to enhance their knowledge in research, engineering, and other tangential skills pertinent to the aerospace industry as well as the operation of a large, engineering organization. SOAR is a professional and competitive research organization that produces industry-relevant data, innovations, and reports.



Values, Mission and Vision:

At SOAR, ingenuity, teamwork, positive community outreach, and driving passion are all core values that drive each and every member of this organization. These are not only values that our members join with, but they are also values that are further developed throughout a member's involvement with our operations.

Our mission is to continue competing at NSL, advance our research into liquid propulsion rocket engines, and aid our members and the Tampa community to build relevant technical and professional skill sets to contribute to today's society.

We aim to develop the next generation of innovative, problem-solving professionals within the engineering industry and beyond to engineer a better world. We also aim to contribute to the promotion of STEM within the USF and Tampa community through our active outreach programs, for the Sky is Not The Limit.

Why do we Need Sponsors?

Sponsor funding is a vital part of SOAR's operation and achievements. Sponsoring our organization opens up new publicity opportunities to a diverse collegiate audience. With our active involvement in one of the most diverse and largest universities in Florida combined with the support we get from our sponsors, we are able to further develop our competitive and research divisions as well as expand our outreach and community involvement efforts.

By sponsoring SOAR, you gain access to unique publicity opportunities and align your brand with a passionate, driven group of students who are dedicated to advancing the field of aerospace. Your support directly contributes to the professional development of future engineers, helping us reach our full potential as an organization. We invite you to join us on this journey and amplify your positive impact on the community. **Sponsor benefits are highlighted in page 7.**



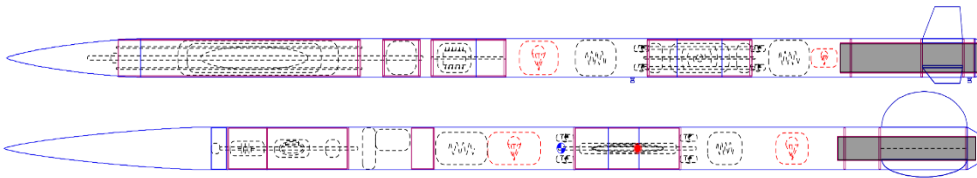
Current Projects

1. NASA Student Launch (NSL):

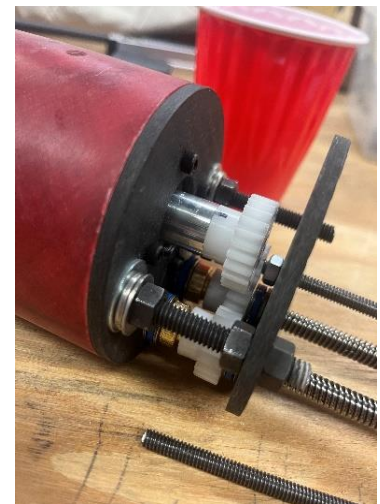
The [NASA Student Launch Initiative](#) challenges university students across the nation to design, analyze, and build a high-powered rocket capable of carrying an innovative payload to an exact, team-selected altitude. The competition requires nearly 1,000 pages of reports and documentation, and three team presentations to NASA prior to launch week. In years prior, SOAR has developed and delivered payloads such as visual odometry algorithms, radio frequency command modules, rover deployment systems, and more.

In the previous school year, 2023-2024, SOAR developed a new payload of an independent drone lander to self-land when launched at apogee and maintain viable human survivability metrics as a simulated landing of astronauts on celestial objects. We have also developed a deployment system that opens the rocket mid-air to jettison the drone out of the rocket, initiating the drone's self-landing sequence at apogee. To pull off such an operation, three teams had to work together: Aerostructures, Payload EE/CS, and Payload Mechanical.

- **Aerostructures:** The aerostructures team is responsible for the design, manufacture, and assembly of the launch vehicle that transports the payload. This includes manufacturing rocket components such as fins as well as running flight simulations and ensuring parachute deployment. The team's goal is to maximize the success of the payload instrument by ensuring a safe flight and recovery.



- **Payload EE/CS:** The payload EE/CS team is responsible for the development of all computer hardware and software systems that both the rocket and the payload need to function properly. Moreover, the team was responsible for the development of a telemetry protocol to make communications between the rocket, the payload, and the ground station.
- **Payload Mechanical:** The payload mechanical team is responsible for the development of all mechanical systems behind the payload deployment and the payload itself. In 2024 the payload mechanical team developed a jettisoning quadcopter drone payload.





2. Torito

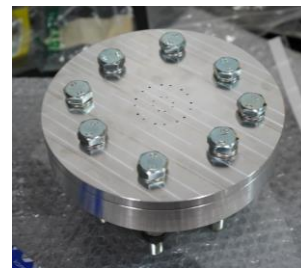
The Torito Liquid Propulsion team is building a liquid bipropellant rocket to compete in the FAR Ten Cents Per Foot (FAR-TCPF) Challenge. The goal in this competition is to launch a liquid propulsion rocket using nitrous oxide as high as possible. The engine of the propulsion rocket will be entirely student researched and built. We expect that Torito's rocket will likely reach an estimated altitude of 10,000 ft. So far, the team has focused on propulsion system development. Torito has built a test stand, developed an engine design, begun data acquisition systems development, and manufactured engine components. Once the propulsion system is fully developed and successfully tested, rocket integration will begin. To develop the propulsion system, Torito consists of four sub teams:



- **Propulsion:** This sub team is responsible for developing a feed system, engine design, plumbing, and integration
- **Chemical:** This sub team is responsible for analyzing propellant reactions, running simulations such as NASA CEA, rocket propulsion analysis, and computational fluid dynamics, and performing the necessary thermodynamic and fluid dynamic calculations
- **Electrical:** This sub team is responsible for developing data acquisition software and electrical systems so that temperature, thrust, and pressure values can be measured from the propulsion system
- **Structures:** This sub team is responsible for creating a test stand and integrating the propulsion system onto it so that the engine can be tested and run successfully before being incorporated into a rocket



Torito's Engine Test Stand



Recent Propellant Injector Plate

3. Outreach and Professional Development

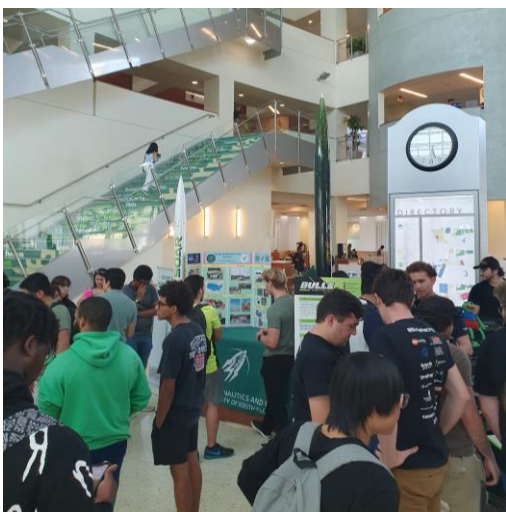
At the Society of Aeronautics and Rocketry (SOAR) at USF, we are dedicated to inspiring the next generation of engineers. We actively participate in events like the Great American Teach-In, where we introduce local students to aerospace through hands-on demonstrations and career insights. Additionally, we play a key role in the USF Student Organization Showcases and the USF Engineering Expo, where we engage with prospective students and the community, showcasing our projects and fostering interest in engineering.

Our commitment extends beyond these events through our participation in the NASA University Student Launch Initiative (USLI), which requires us to meet annual outreach quotas. This ensures that SOAR is not only advancing in technical expertise but also making a tangible impact on the future of the engineering profession by organizing and participating in educational initiatives.



USF Engineering Expo 24'

YOUTH IMPACTED
FROM 2017-2023
+3,900



Racing & Rocketry Exhibition



Patel Elementary School

Member Benefits

1- Project Involvement:

One of the core benefits that our members get is the active Project Involvement: Whether its developing embedded or mechanical systems for NSL's payload, aerostructure design, research into liquid propulsion technology, or designing hardware and software for the rocket and payload, members will learn and apply varying skills.

SOAR Team	Typical Skills Gained by Members
Torito Team	• NASA Chemical Equalizer Analysis • Computational Fluid Dynamics • Rocket Propulsion Analysis software • Scientific Documentation & Technical Writing
NSL Mechanical Team	• Computer Aided Design • Design for Assembly and Manufacturing • Design Iteration and Rapid Prototyping • Additive and Subtractive Manufacturing • CNC Programming and Operation
NSL Electrical Engineering /Computer Science Team	• Object Oriented Programming • Circuit design and analysis • Creating state machines • Git • PCB Design & Soldering • Embedded systems

2- Tripoli Rocket Association (TRA) Certification Programs:

Tripoli is a non-profit, international organization dedicated to the education, advancement, and safe operation of amateur high-power rocketry. The TRA recognizes three levels of certification, each allowing the rocketeer to launch increasingly powerful rockets. At SOAR, education is one of our core values. To support our members in their learning journey, we offer **free Level 1 and Level 2 Certification build classes**, where students can gain hands-on experience in rocketry aerostructure and mechanical design as well as safe recovery techniques. Importantly, SOAR fully sponsors these certifications, covering all associated costs to ensure that financial barriers do not hinder our members' educational and professional development.



3- Outreach Initiatives:

SOAR recognizes the importance of developing technical experience as well as strong professional skills. Student members are given opportunities to develop and hone their professional skills through tabling events, presentations in different technical classes and SOAR-sponsored workshops, and events in collaboration with other student organizations. Partnerships with societies such as the Society of Asian Scientists and Engineers (SASE) and the Society of Hispanic Professional Engineers (SHPE) provide resume-building events, networking socials, and other amazing opportunities to develop strong professional candidates.



Since January 2024 we have:



- Increased our Instagram User Reach by %10+ to have a 3.3K user reach so far
- Reached 1.5K+ followers
- Maintained 2.4k Profile Visits



%10+
User Reach



1.5k+
Followers



+2.4K
Profile Visits

Our Next Steps

NSL:

The next step for NSL is to compete again in the 2024-2025 Competition. While we await the kick off announcement of the next iteration of NSL in Fall 2024, the team is currently developing an ABS (Active Braking System) to integrate into the next rocket so that the team will hit their target apogee (maximum altitude). Achieving the target apogee is 10% of the scoring criteria for the NSL competition, so an ABS could potentially raise the team rank significantly next year.

Torito:

The immediate next steps for Torito are to integrate the engine design onto the test stand to complete a cold flow test. After that, the liquid propellants will be added to complete a static fire test. Finally, the engine will be removed from the test stand and integrated into the rocket to complete a successful launch at the Far Ten Cents Per Foot challenge launch site.

Sponsor Benefits

The following benefits are valid for the academic year of sponsorship (2024-2025). All Donations are tax deductible. Of course, alternative requests can always be considered. Both monetary and in-kind (material) gifts are greatly appreciated.

	SOAR Starter \$1000+	SOAR Premium \$2000+	SOAR Premium+ \$7000+	SOAR Platinum \$8500+
Sponsor's Logo Featured on Website	✖	✖	✖	✖
Sponsor's Logo Featured on all Social Media Posts by SOAR	✖	✖	✖	✖
Sponsor's Logo Featured on All Presentations and Videos		✖	✖	✖
Sponsor's Logo or Name on all Apparel Ordered in Year of Sponsorship		✖	✖	✖
Sponsor's Small Logo on all rockets and products developed by SOAR the year of sponsorship			✖	✖
Sponsor's Large Logo on all Products Developed by all SOAR the Year of Sponsorship			✖	✖
Sponsor's Logo on all posters/banners of all of SOAR's Events, Exhibitions, and Career Fairs				✖
Access To SOAR Members' Resume Database				✖
Invitation to Host One Career Fair/Recruitment Event & One Workshop at USF				✖

Contact Us:

Thank you for your interest! If you have any questions or want to learn more about the USF Society of Aeronautics and Rocketry before contributing, please contact us directly and/ or follow us on social media using the QR Code below:

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SOAR Socials



If you have already decided to contribute directly to SOAR now, you can easily do so at <http://giving.usf.edu/online>. Be sure to select fund **#220111** when donating. Thank you!

